DENSE OPEN OF A FINITE PRESENTED SPACE OVER A NOETHERIAN BASE IMMERSES IN AFFINE SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3313-3366, following the statement at 3305-3311.

Editorial note. The single '(details omitted)' covers the amalgamated immersion A^n_X coprod A^m_X -> A^{max(n,m)+1}_X, a standard construction independent of the main argument; flagged in the notes.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 3331 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Noetherianity of finite presentation morphisms. Morphisms of Spaces, Lemma spaces-morphisms-lemma-finite-presentation-noetherian (cross-chapter, cited at ~3314-3316). Y is Noetherian.
- Dense open subscheme of a quasi-separated space. Properties of Spaces, Proposition spaces-properties-proposition-locally-quasi-separated-open-dense-scheme (cross-chapter, cited at ~3317-3319 and ~3359-3360). Reduces Y (resp. X) to a scheme.
- Noetherian topology decomposition. Topology, Lemma topology-lemma-Noetherian and Properties, Lemma properties-lemma-Noetherian-topology (cross-chapter, cited at ~3321-3323). Remove intersections of irreducible components.
- Soberness of quasi-separated spaces. Properties of Spaces, Lemma spaces-properties-lemma-quasi-separated-sober (cross-chapter, cited at ~3336-3338). T is irreducible with a unique generic point, the image of the generic point of Y.
- Reduced induced closed subspace. Properties of Spaces, Definition spaces-properties-definition-reduced-induced-space (cross-chapter, cited at ~3341-3344). The quasi-coherent ideal I cutting out T.
- Quasi-affine factorization. Morphisms, Lemma morphisms-lemma-quasi-affine-finite-type-over-S (cross-chapter, cited at ~3362-3364). Factors V -> U as V -> A^n_U -> U.

Source reference. Original reference 04II: focused-seeds/geometry/S9-spaces-limits-embedding-into-affine-over-qs/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 01OZ: focused-seeds/geometry/S9-spaces-limits-embedding-into-affine-over-qs/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04ZL: focused-seeds/geometry/S9-spaces-limits-embedding-into-affine-over-qs/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 047X: focused-seeds/geometry/S9-spaces-limits-embedding-into-affine-over-qs/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06NJ: focused-seeds/geometry/S9-spaces-limits-embedding-into-affine-over-qs/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06NH: focused-seeds/geometry/S9-spaces-limits-embedding-into-affine-over-qs/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0052: focused-seeds/geometry/S9-spaces-limits-embedding-into-affine-over-qs/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. Assume X Noetherian and f of finite presentation. Then there exists a dense open $V \subset Y$ and an immersion $V \to \mathbf{A}^n_X$.*

Proof. The assumptions imply that Y is Noetherian (Morphisms of Spaces, Lemma 04ZL). Then Y is quasi-separated, hence has a dense open subscheme (Properties of Spaces, Proposition 06NH). Thus we may assume that Y is a Noetherian scheme. By removing intersections of irreducible components of Y (use Topology, Lemma 0052 and Properties, Lemma 01OZ) we may assume that Y is a disjoint union of irreducible Noetherian schemes. Since there is an immersion

$$\mathbf{A}^n_X \amalg \mathbf{A}^m_X \longrightarrow \mathbf{A}^{\max(n,m)+1}_X$$

(details omitted) we see that it suffices to prove the result in case Y is irreducible.

Assume Y is an irreducible scheme. Let $T \subset |X|$ be the closure of the image of $f : Y \to X$. Note that since $|Y|$ and $|X|$ are sober topological spaces (Properties of Spaces, Lemma 06NJ) T is irreducible with a unique generic point ξ which is

the image of the generic point η of Y. Let $\mathcal{I} \subset X$ be a quasi-coherent sheaf of ideals cutting out the reduced induced space structure on T (Properties of Spaces, Definition 047X). Since $\mathcal{O}_{Y,\eta}$ is an Artinian local ring we see that for some $n > 0$ we have $f^{-1}\mathcal{I}^n\mathcal{O}_{Y,\eta} = 0$. As $f^{-1}\mathcal{I}\mathcal{O}_Y$ is a finite type quasi-coherent ideal we conclude that $f^{-1}\mathcal{I}^n\mathcal{O}_V = 0$ for some nonempty open $V \subset Y$. Let $Z \subset X$ be the closed subspace cut out by $\mathcal{I}^n$. By construction $V \to Y \to X$ factors through Z. Because $\mathbf{A}^n_Z \to \mathbf{A}^n_X$ is an immersion, we may replace X by Z and Y by V. Hence we reach the situation where Y and X are irreducible and $Y \to X$ maps the generic point of Y onto the generic point of X.

Assume Y and X are irreducible, Y is a scheme, and $Y \to X$ maps the generic point of Y onto the generic point of X. By Properties of Spaces, Proposition 06NH X has a dense open subscheme $U \subset X$. Choose a nonempty affine open $V \subset Y$ whose image in X is contained in U. By Morphisms, Lemma 04II we may factor $V \to U$ as $V \to \mathbf{A}^n_U \to U$. Composing with $\mathbf{A}^n_U \to \mathbf{A}^n_X$ we obtain the desired immersion. □